Differentiable plasticity FEM in JAX

what a mechanical test sees, and how its derivatives flow

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These notes build a small, fully differentiable finite-element model of a punch test on an elastic–plastic plate, and use it to follow how a change in the hardening law travels through the computation to the measured load. The model was written on core JAX; the figures come from it or are drawn in TikZ. The material, geometry and parameter values are generic and chosen for teaching.

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1 What a mechanical test sees

A test imposes boundary data (a displacement, a traction, contact with a fixture) and a loading path. The body responds with fields $u(x,t)$, $\varepsilon(x,t)$, $\sigma(x,t)$ and internal variables $q(x,t)$. We observe only a compressed record: here the punch load at each depth, $y = \{L(d_k)\}_{k=1}^{N_d}$. The constitutive law is never measured directly. It reaches the data only after passing through the local plastic response, equilibrium with contact, the loading history and the readout (Fig. 1). Calibration is therefore an inverse problem,

$$y_{\text{obs}} = \mathcal{G}(\theta) + \epsilon, \tag{1}$$

and the derivative $\partial\mathcal{G}/\partial\theta$ is the object that tells us what the test can say about θ.

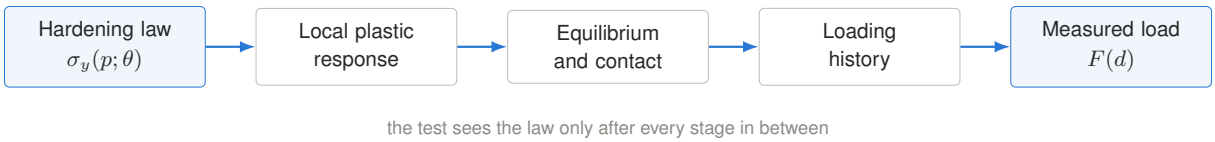


Figure 1: The chain from the hardening law to the measured load.

2 J_2 plasticity with combined hardening

With $s = \operatorname{dev}\sigma$, back stress X and accumulated plastic strain p,

$$f = \sigma_{eq}(s - X) - \sigma_y(p) \le 0, \qquad \sigma_{eq}(\xi) = \sqrt{\tfrac{3}{2}\,\xi : \xi}, \tag{2}$$

$$\dot{\varepsilon}^p = \dot{\gamma}\, n, \quad n = \frac{3}{2}\,\frac{\xi}{\sigma_{eq}(\xi)}, \qquad \dot{p} = \dot{\gamma}, \tag{3}$$

$$\sigma_y(p) = \sigma_0 + Q\,(1 - e^{-\beta p}) \qquad \text{(Voce, isotropic)}, \tag{4}$$

$$\dot{X} = \tfrac{2}{3} C\,\dot{\varepsilon}^p - \gamma_k\,\dot{p}\,X \qquad \text{(Armstrong–Frederick, kinematic)}. \tag{5}$$

Isotropic hardening grows the elastic domain; kinematic hardening moves it (Fig. 2). Under a load reversal the second gives early reverse yielding, the Bauschinger effect (Fig. 3). Positivity of $\sigma_y(0)$ and $\sigma_y' \ge 0$ keep the incremental problem well posed. In tabulated form (as in commercial codes) σ_y is simply a function of p; here we use the three-parameter Voce chart $\theta = (\sigma_0, Q, \beta)$.

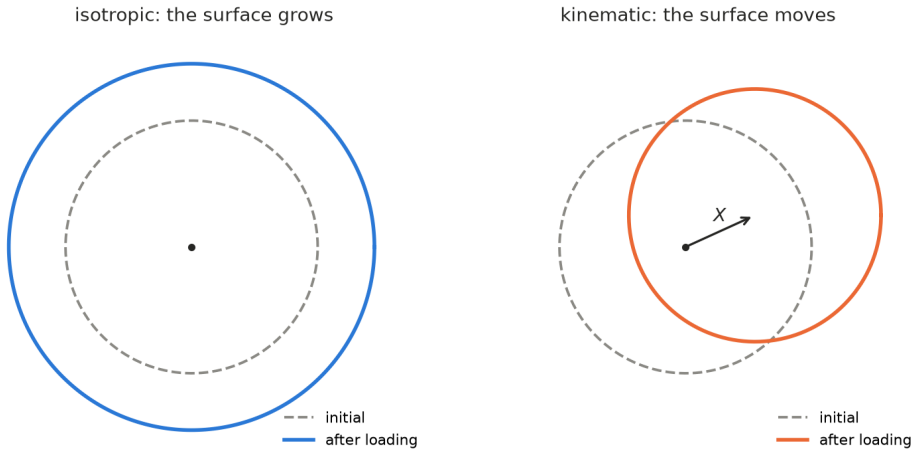


Figure 2: Deviatoric-plane view: isotropic hardening dilates the surface, kinematic hardening translates it by X.

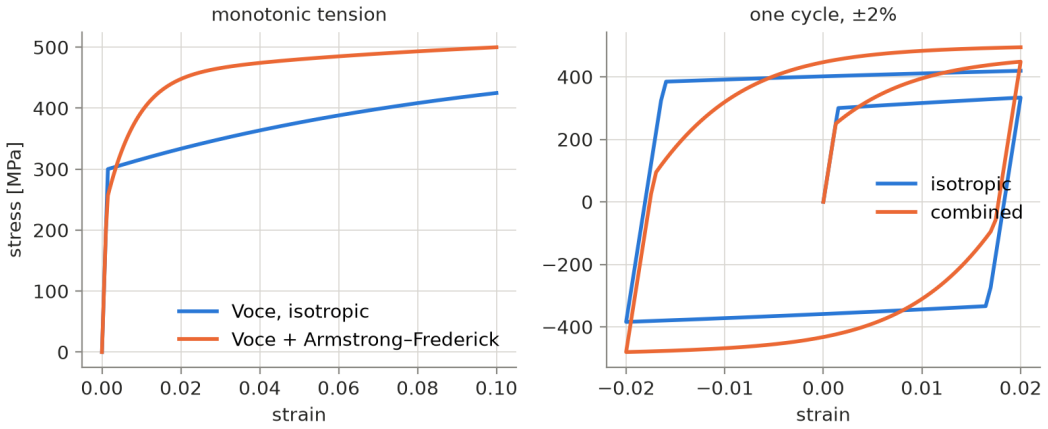


Figure 3: Monotonic tension and one strain cycle for pure Voce and for Voce + Armstrong–Frederick.

3 The return map and its Jacobian

Backward Euler over an increment, with local unknowns $v = (\sigma, \varepsilon^p, X, p)$, gives the local residual $h(v) = 0$ (Fig. 4): elastic update, flow rule, back-stress evolution, accumulation and yield consistency. Starting from the trial state $s^{\mathrm{tr}} = 2G\,\mathrm{dev}(\varepsilon - \varepsilon_n^p)$, everything collapses to one scalar equation for $\Delta\gamma$:

$$g(\Delta\gamma) = \Big\| s^{\mathrm{tr}} - \frac{X_n}{1+\gamma_k\Delta\gamma} \Big\| - \Big(2G\sqrt{\tfrac{3}{2}} + \sqrt{\tfrac{2}{3}}\,\frac{C}{1+\gamma_k\Delta\gamma}\Big)\Delta\gamma - \sqrt{\tfrac{2}{3}}\,\sigma_y(p_n + \Delta\gamma) = 0. \tag{6}$$

In code this root is found with Newton inside `jax.lax.custom_root`. The implicit function theorem then gives every derivative of the result without differentiating through the iterations:

$$\frac{d\Delta\gamma}{d(\cdot)} = -\Big(\frac{\partial g}{\partial \Delta\gamma}\Big)^{-1} \frac{\partial g}{\partial (\cdot)}. \tag{7}$$

The algorithmic (consistent) tangent $\mathbb{D} = \partial\sigma_{n+1}/\partial\varepsilon_{n+1}$ is then a single `jax.jacfwd`. For pure isotropic hardening it agrees with the textbook closed form

$$\mathbb{D} = K\,\mathbf{1}\otimes\mathbf{1} + 2G\theta_1\,\mathbb{I}_{\mathrm{dev}} - 2G\theta_2\,\hat{n}\otimes\hat{n}, \qquad \theta_1 = 1 - \frac{3G\Delta\gamma}{\sigma_{eq}^{\mathrm{tr}}}, \quad \theta_2 = \frac{1}{1+H/3G} - (1-\theta_1), \tag{8}$$

to round-off, and it needs no change for the combined model.

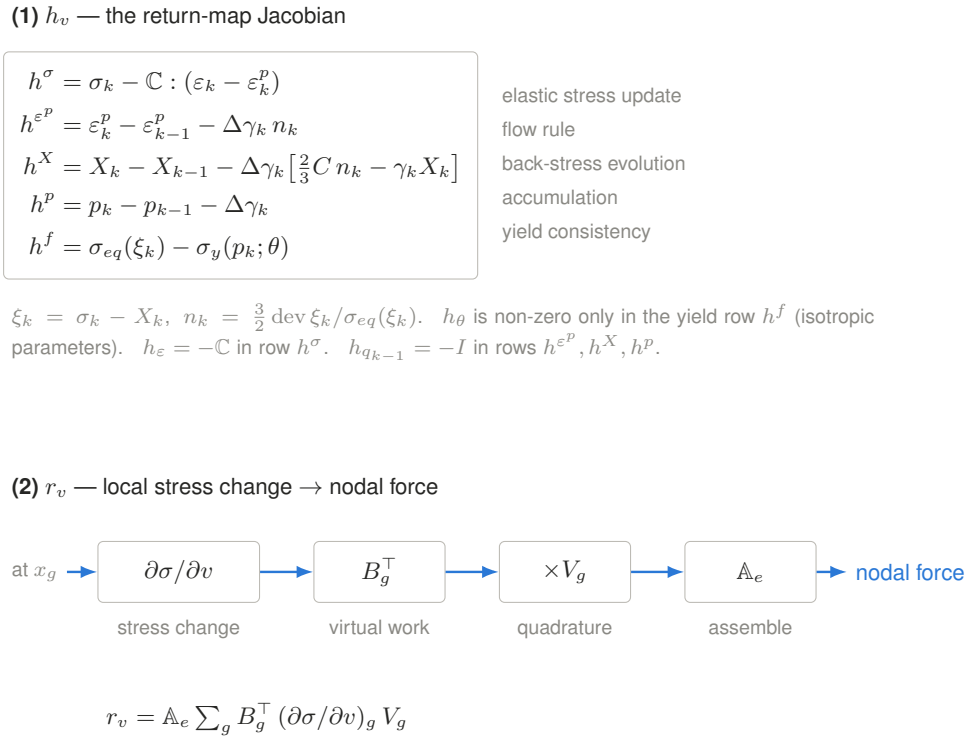


Figure 4: The local residual and the map from a local stress change to nodal force.

4 The incremental finite-element problem

Per increment three residuals are coupled (Fig. 5): the local constitutive residual h_k at every Gauss point, global equilibrium r_k, and the contact conditions. Newton solves the linearized saddle problem; condensing the local solve into the global one gives the consistent tangent

$$K_k = r_U - r_v\, h_v^{-1}\, h_U, \tag{9}$$

the material's response to nodal motion. Because K_k is exact, Newton converges quadratically once the contact set settles.

The teaching model is an axisymmetric clamped plate (radius 2 mm, thickness 0.5 mm) pressed by a rigid ball (radius 1.25 mm). It uses 4-node quads with 2×2 Gauss points, small strain, and frictionless penalty contact, so the saddle reduces to $K_k\, \delta U = -r_k$ with the penalty stiffness inside K_k. The pieces map one-to-one onto JAX (Fig. 7): the internal force is the transpose of the strain map (`jax.vjp`), the tangent is `jax.jacfwd` of the residual, and the global Newton also sits in `custom_root`, so $\partial U/\partial(\cdot) = -K^{-1}\partial R/\partial(\cdot)$ is applied directly.

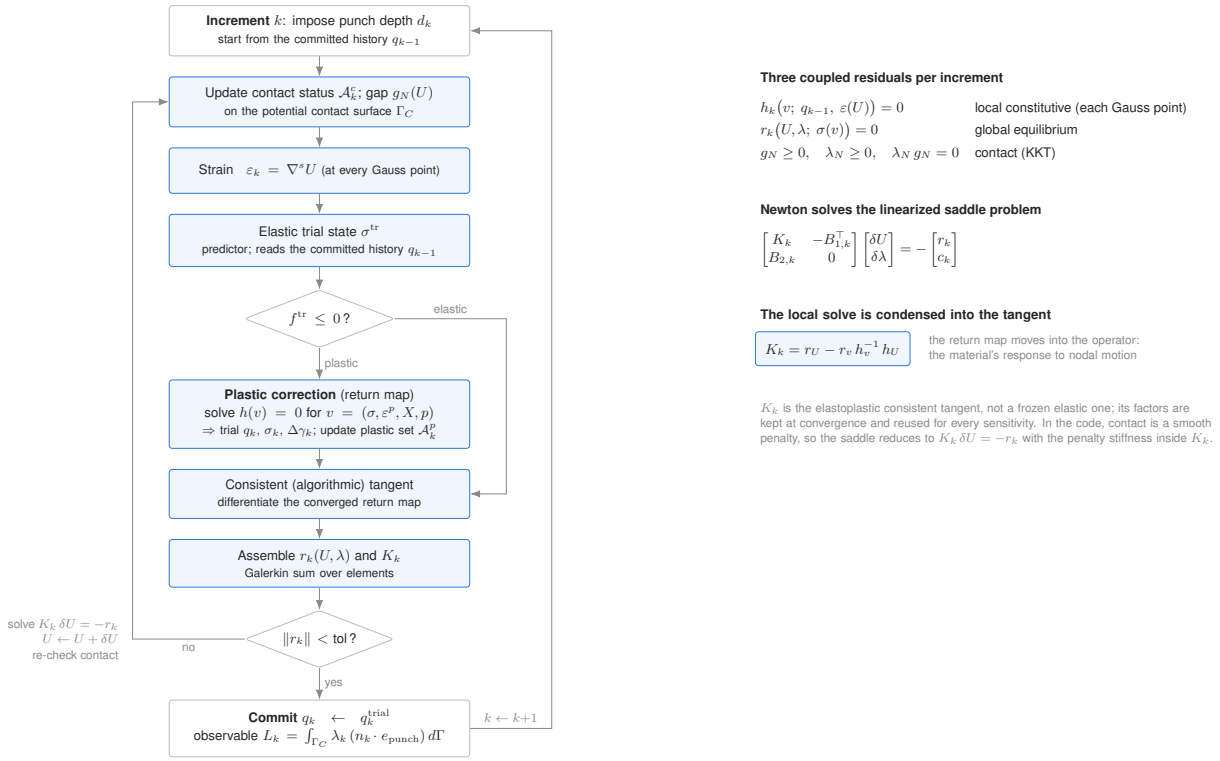


Figure 5: The forward solve for one increment, with the residuals it balances and the condensed tangent.

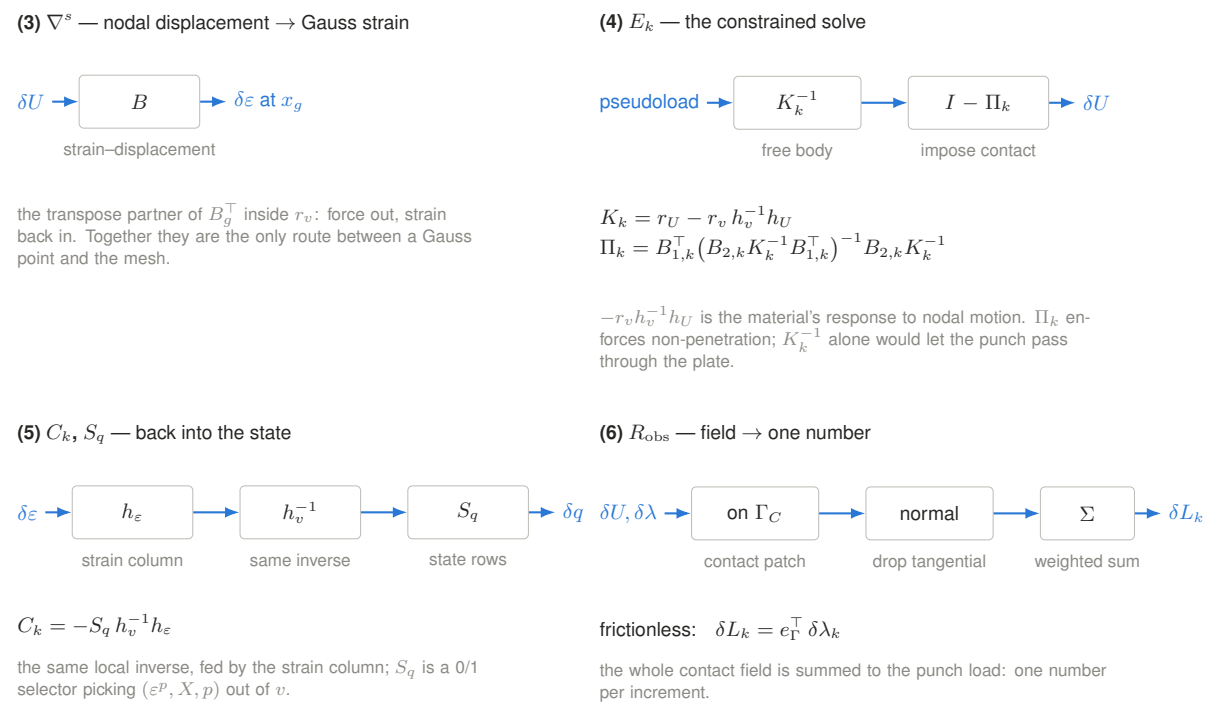


Figure 6: Kinematics, the constrained solve, the state update and the readout.

step	JAX tool
Every Gauss point	`jax.vmap`
Plastic multiplier $\Delta\gamma$	`jax.lax.custom_root`
Consistent tangent K	`jax.jacfwd(residual)`
Internal force $B^\top\sigma$	`jax.vjp(strain)`
Global Newton	`jax.lax.custom_root`
Load increments	`jax.lax.scan`
$dF/d\theta$, one pass per parameter	`jax.jacfwd`
$dF/d\theta$, one pass per output	`jax.jacrev`

Figure 7: Which JAX tool does which job.

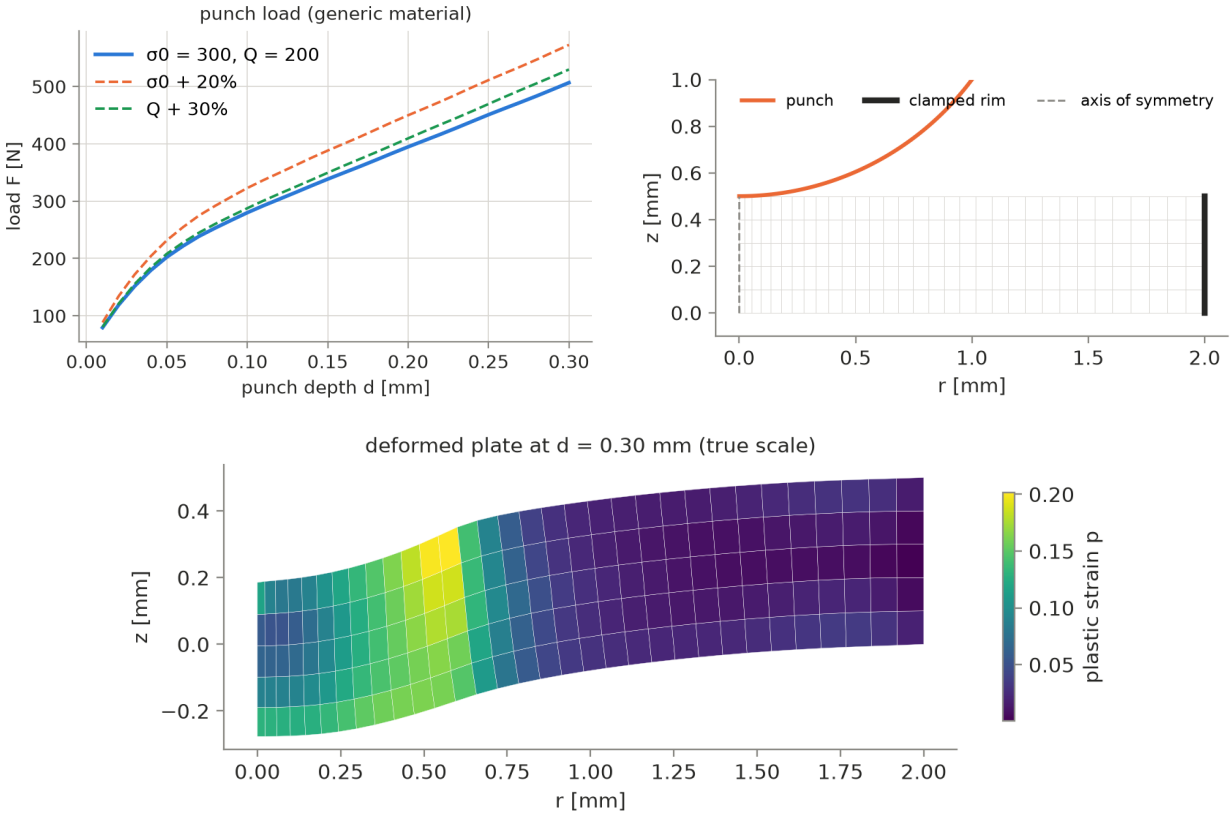


Figure 8: Load–depth curves (with two parameter changes), the mesh, and accumulated plastic strain at the last increment.

5 Linearizing the forward model

A parameter change enters in one place: the yield row, evaluated at the plastic strain reached at each plastic Gauss point,

$$\frac{\partial h^f}{\partial \theta}\Big|_g = -\frac{\partial \sigma_y}{\partial \theta}(p_k(x_g);\theta). \tag{10}$$

From there it is *resisted* by the local block h_v^{-1}, *redistributed* by equilibrium under contact (the same factorized K_k as the forward Newton, one extra right-hand side), *carried* to later increments through the committed state, and *read out* as a change in load (Fig. 9). With $s_k = \partial q_k/\partial\theta$, the pseudoload and the state recursion are

$$f_{H,k} = r_v\, h_v^{-1}\big(h_{q_{k-1}} s_{k-1} + h_\theta\big) + f_{\Gamma,k}, \tag{11}$$

$$s_k = -S_q\, h_v^{-1}\big(h_\varepsilon B\, U_{\theta,k} + h_{q_{k-1}} s_{k-1} + h_\theta\big). \tag{12}$$

Writing $s_k = \Phi_k s_{k-1} + \Upsilon_k\, \delta\theta$ and unrolling (Fig. 10) shows that every increment injects, and the number of carry steps on a path is how old that injection is. Assembled, the Jacobian factors as

$$J = \mathcal{D} + \mathcal{H} \circ (\mathrm{Id} - \mathcal{T})^{-1} \circ \mathcal{G}, \qquad (\mathrm{Id} - \mathcal{T})^{-1} = \textstyle\sum_{n=0}^{N-1} \mathcal{T}^n, \tag{13}$$

a finite Neumann series because the history transfer $\mathcal{T}$ is strictly causal (Fig. 11). Inside each operator the same two paths appear, one ending in the state and one in the load (Fig. 12).

Direct or adjoint. Direct evaluation fills J column by column (one extra solve per parameter per increment); adjoint fills it row by row (one backward sweep per output). With three parameters and thirty depths, direct is cheaper; with many parameters and a scalar objective, adjoint is. In JAX these are `jacfwd` and `jacrev`. Both were checked, together with finite differences and a hand-written direct recursion, against each other (Fig. 13).

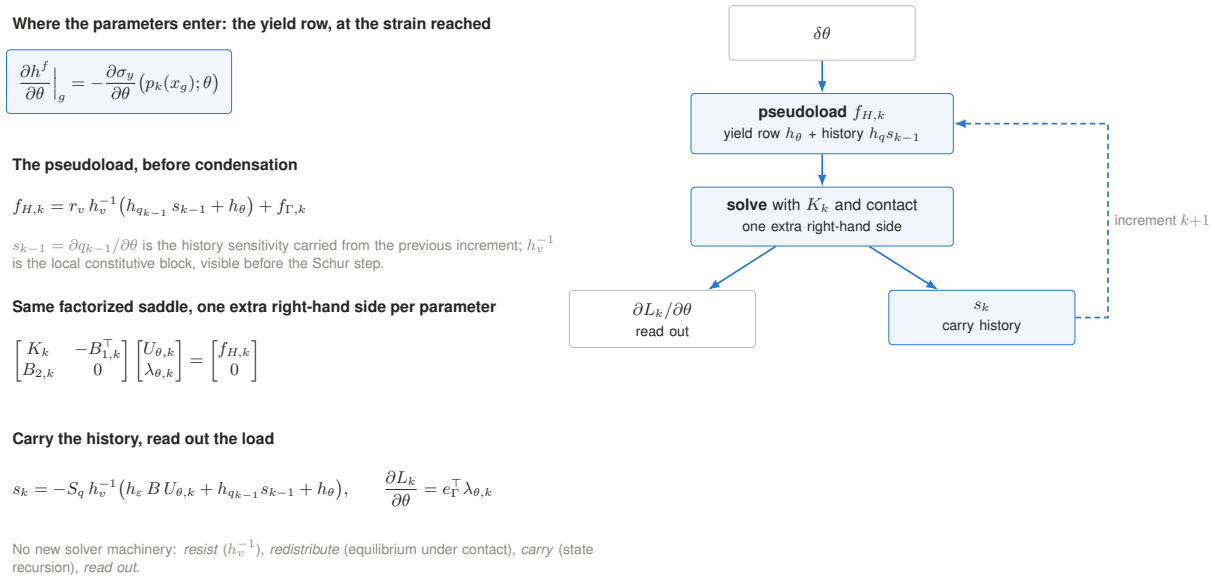


Figure 9: Sensitivity rides the same structure as the forward solve.

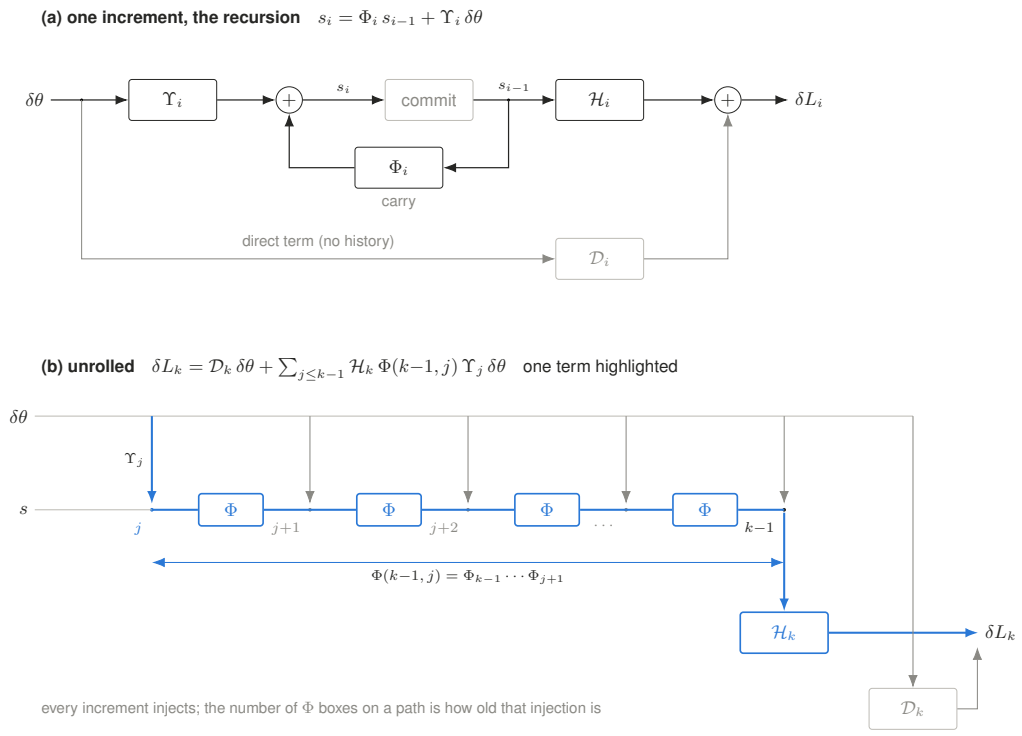


Figure 10: (a) The recursion for one increment. (b) Unrolled over increments, one injection highlighted.

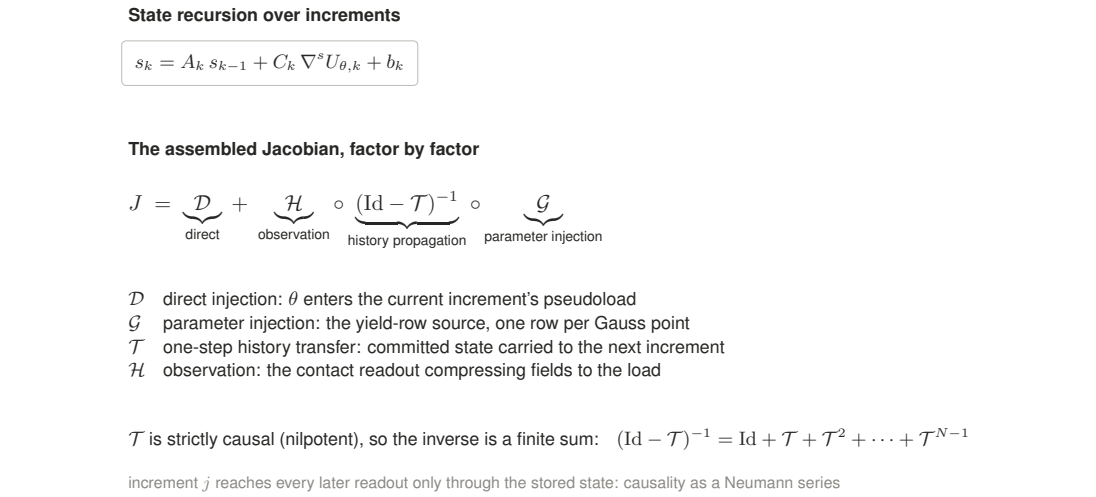
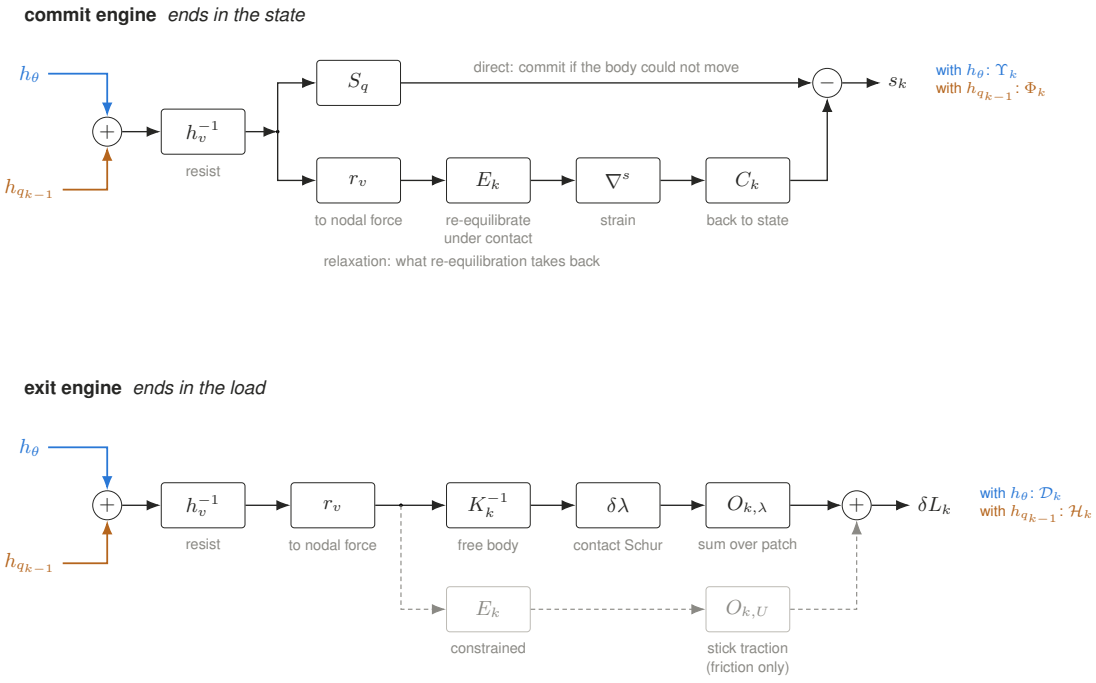


Figure 11: The assembled Jacobian, factor by factor.



The two sources add: with only h_θ the composite is Υ_k, with only $h_{q_{k-1}}$ it is Φ_k; a real increment has both. The commit engine carries the minus and ends in the state; the exit engine carries the plus and ends in the load. Both are the same factorized saddle solve, read two ways.

Figure 12: Inside the operators: the state path and the load path, each fed by the parameter source and the history source.

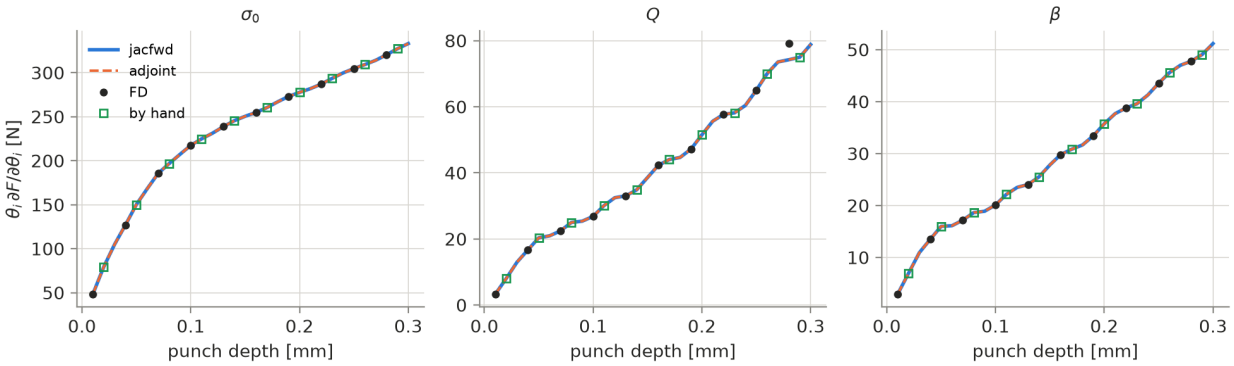


Figure 13: $\theta_i\, \partial L/\partial \theta_i$ for the punch: forward mode, adjoint, finite differences and the hand-written recursion agree.

6 A local Bayesian view with three parameters

Linearize at θ_0: $\mathcal{G}(\theta_0 + \delta\theta) \approx \mathcal{G}(\theta_0) + J\,\delta\theta$, with noise $\epsilon \sim \mathcal{N}(0, \Sigma)$. Then

$$\mathcal{J}(\theta) = \tfrac{1}{2}\|y - \mathcal{G}(\theta)\|^2_{\Sigma^{-1}} + \tfrac{1}{2}\|\theta - \theta_{\mathrm{pr}}\|^2_{Q_{\mathrm{pr}}}, \qquad Q_{\mathrm{post}} \approx Q_{\mathrm{pr}} + \Gamma, \qquad \Gamma = J^\top \Sigma^{-1} J. \tag{14}$$

Γ (the Fisher information, or Gauss–Newton Hessian) carries everything the experiment contributes locally. Its generalized eigenpairs $\Gamma\psi_i = \lambda_i\, Q_{\mathrm{pr}}\psi_i$ give the directions the data inform: along mode i the posterior variance is the prior variance times $1/(1+\lambda_i)$, so modes with $\lambda_i > 1$ have their variance at least halved. Comparing Γ for tension and for the punch (Fig. 14) shows which combinations of (σ_0, Q, β) each test pins down. All of this is local: one anchor, one noise model, one three-parameter chart.

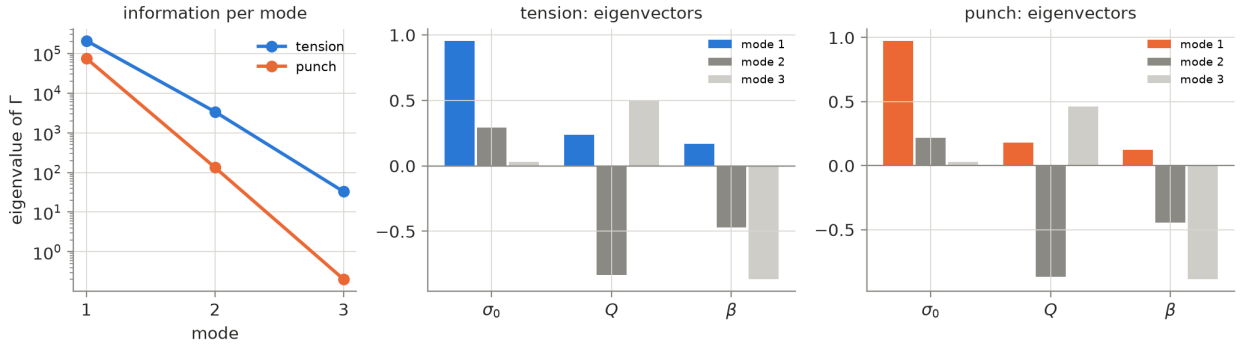


Figure 14: Eigenvalues and eigenvectors of Γ in log-parameters, 1% noise relative to each test's peak.

7 Where this leads

The same machinery can be asked a sharper question: not "how does the load change with three constants" but "where in plastic-strain space, at each depth, is the load sensitive to the hardening law at all". Figure 15 is a first look (work in progress).

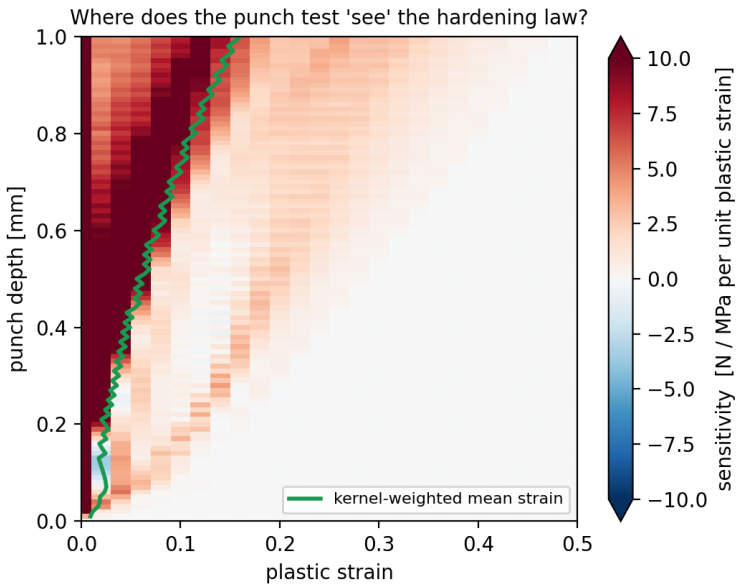


Figure 15: Work in progress: sensitivity of the punch load to the hardening law, resolved in plastic strain, with the kernel-weighted mean strain.

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